Chapter 11

REQUISITE VARIETY

11/1. In the previous chapter we considered regulation from the biological point of view, taking it as something sufficiently well understood. In this chapter we shall examine the process of regulation itself, with the aim of finding out exactly what is involved and implied. In particular we shall develop ways of *measuring* the amount or degree of regulation achieved, and we shall show that this amount has an upper limit.

11/2. The subject of regulation is very wide in its applications, covering as it does most of the activities in physiology, sociology, ecology, economics, and much of the activities in almost every branch of science and life. Further, the types of regulator that exist are almost bewildering in their variety. One way of treating the subject would be to deal seriatim with the various types, and chapter 12 will, in fact, indicate them. In this chapter, however, we shall be attempting to get at the core of the subject—to find what is common to all.

What is common to all regulators, however, is not, at first sight much like any particular form. We will therefore start anew in the next section, making no explicit reference to what has gone before. Only after the new subject has been sufficiently developed will we beam to consider any relation it may have to regulation.

11/3. *Play and outcome.* Let us therefore forget all about regulation and simply suppose that we are watching two players, *R* and *D,* who are engaged in a game. We shall follow the fortunes of *R,* who is attempting to score an *a.* The rules are as follows. They have before them Table 11/3/1, which can be seen by both:

Table 11/3/l

		R		
		α	β	γ
	1	*b*	*a*	*c*
D	2	*a*	*c*	*b*
	3	*c*	*b*	*a*

D must play first, by selecting a number, and thus a particular row. *R*, knowing this number, then selects a Greek letter, and thus a particular column. The italic letter specified by the intersection of the row and column is the **outcome.** If it is an *a, R* wins; if not, *R* loses.

Examination of the table soon shows that with this particular table *R* can win always. Whatever value *D* selects first, *R* can always select a Greek letter that will give the desired outcome. Thus if *D* selects 1, *R* selects β; if *D* selects 2, *R* selects α*;* and so on. In fact, if *R* acts according to the transformation

↓	1	2	3
	β	α	γ

then he can always force the outcome to be *a.*

R's position, with this particular table, is peculiarly favourable, for not only can *R* always force *a* as the outcome, but he can as readily force, if desired, *b* or c as the outcome. *R* has, in fact, complete control of the outcome.

Ex. 1: What transformation should *R* use to force *c* as outcome?

Ex. 2: If both *R's* and *D's* values are integers, and the outcome *E* is also an integer, given by

$$E = R - 2D,$$

find an expression to give *R* in terms of *D* when the desired outcome is 37.

Ex. 3: *A* car's back wheels are skidding. *D* is the variable "Side to which the tail is moving", with two values, Right and Left. *R* is the driver's action "Direction in which he turns the steering wheel" with two values, Right and Left. Form the 2 x 2 table and fill in the outcomes.

Ex. 4: If *R's* play is determined by *D's* in accordance with the transformation

↓	1	2	3
	β	α	γ

and many games are observed, what will be the variety in the many outcomes?

Ex. 5: Has *R* complete control of the outcome if the table is triunique?

11/4. The Table used above is, of course, peculiarly favourable to *R.* Other Tables are, however, possible. Thus, suppose *D* and *R,* playing on the same rules, are now given Table 11/4/1 in which *D* now has a choice of five, and *R* a choice of four moves.

If *a* is the target, *R* can always win. In fact, if *D* selects 3, *R* has several ways of winning. As every row has at least one *a, R* can always force the appearance of *a* as the outcome. On the other hand, if the target is *b* he cannot always win. For if *D* selects 3, there is no move by *R* that will give *b* as the outcome. And if the target is *c, R* is quite helpless, for *D* wins always.

It will be seen that different arrangements within the table, and different numbers of states available to *D* and *R,* can give rise to a variety of situations from the point of view of *R.*

Table 11/4/l

		R			
		α	β	γ	δ
	1	*b*	*d*	*a*	*a*
	2	*a*	*d*	*a*	*d*
D	3	*d*	*a*	*a*	*a*
	4	*d*	*b*	*a*	*b*
	5	*d*	*a*	*b*	*d*

Ex. 1: With Table I l /4/ l, can *R* always win if the target is *d*?

Ex. 2: (Continued.) What transformation should *R* use?

Ex. 3: (Continued.) If a is the target and *D,* for some reason, never plays 5, how can *R* simplify his method of play?

Ex. 4: A guest is coming to dinner, but the butler does not know who. He knows only that it may be Mr. A, who drinks only sherry or wine, Mrs. B, who drinks only gin or brandy, or Mr. C, who drinks only red wine, brandy or sherry. In the cellar he finds he has only whisky, gin, and sherry. Can he find something acceptable to the guest, whoever comes ?

11/5. Can any *general* statement be made about *R's* modes of play and prospects of success ?

If full generality is allowed in the Table, the possibilities are so many, arbitrary and complicated that little can be said. There is one type, however, that allows a precise statement and is at the same time sufficiently general to be of interest. (It is also fundamental in the theory of regulation.)

From all possible tables let us eliminate those that make *R's* game too easy to be of interest. Ex. 11/4/3 showed that if a column contains repetitions, *R's* play need not be discriminating; that is, *R* need not change his move with each change of *D's* move. Let us consider, then, only those tables in which *no column contains a repeated outcome.* When this is so *R* must select his move on *full* knowledge of *D's* move; i.e. any change of *D's* move must require a change on *R's* part. (Nothing is assumed here about how the outcomes in one column are related to those in another, so these relations are unrestricted.) Such a Table is 11/5/1. Now, some target being given, let *R* specify what his move will be for each move by *D.* What is essential is that, win or lose, he must specify one and

Table 11/5/l

		R		
		α	β	γ
	1	*f*	*f*	*k*
	2	*k*	*e*	*f*
	3	*m*	*k*	*a*
	4	*b*	*b*	*b*
D	5	*c*	*q*	*c*
	6	*h*	*h*	*m*
	7	*j*	*d*	*d*
	8	*a*	*p*	*j*
	9	*l*	*n*	*h*

only one move in response to each possible move of *D.* His specification, or "strategy" as it might be called, might appear:

If *D* selects 1, I shall select γ
„ „ „ 2, „ „ „ α
„ „ „ 3, „ „ „ β
… …
„ „ „ 9, „ „ „ α

He is, of course, specifying a transformation (which must be singlevalued, as *R* may not make two moves simultaneously):

$$\downarrow \begin{matrix} 1 & 2 & 3 & \dots & 9 \\ \gamma & \alpha & \beta & \dots & \alpha \end{matrix}$$

This transformation uniquely specifies a set of outcomes—those that will actually occur if *D,* over a sequence of plays, includes every possible move at least once. For 1 and γ give the outcome *k*, and so on, leading to the transformation:

$$\downarrow \begin{matrix} (1,\gamma) & (2,\alpha) & (3,\beta) & \dots & (9,\alpha) \\ k & k & k & \dots & l \end{matrix}$$

It can now be stated that the variety in this set of outcomes cannot be less than

$$\frac{D\text{'s variety}}{R\text{'s variety}}$$

i.e., in this case, 9/3.

It is easily proved. Suppose *R* marks one element in each row and concentrates simply on keeping the variety of the marked ele-

ments as small as possible (ignoring for the moment any idea of a target). He marks an element in the first row. In the second row he must change to a new column if he is not to increase the variety by adding a new, different, element; for in the initially selected column the elements are all different, by hypothesis. To keep the variety down to one element he must change to a new column at each row. (This is the *best* he can do; it may be that change from column to column is not sufficient to keep the variety down to one element, but this is irrelevant, for we are interested only in what is the least possible variety, assuming that everything falls as favourably as possible). So if *R* has *n* moves available (three in the example), at the n-th row all the columns are used, so one of the columns must be used again for the next row, and a new outcome *must* be allowed into the set of outcomes. Thus in Table 11/5/1, selection of the k's in the first three rows will enable the variety to be kept to one element, but at the fourth row a second element *must* be allowed into the set of outcomes.

In general: If no two elements in the same column are equal, and if a set of outcomes is selected by *R,* one from each row, and if the table has *r* rows and c columns, then *the variety in the selected set of outcomes cannot be fewer than r/c.*

THE LAW OF REQUISITE VARIETY

11/6. We can now look at this game (still with the restriction that no element may be repeated in a column) from a slightly different point of view. If *R, S* move is unvarying, so that he produces the same move, whatever *D, S* move, then *the variety in the outcomes will be as large as the variety in D'S moves. D* now is, as it were, exerting full control over the outcomes.

If next *R* uses, or has available, two moves, then the variety of the outcomes can be reduced to a half (but not lower). If *R* has three moves, it can be reduced to a third (but not lower); and so on. Thus if the variety in the outcomes is to be reduced to some assigned number, or assigned fraction of *D, S* variety, *R, S* variety *must* be increased to at least the appropriate minimum. *Only variety in R, S moves can force down the variety in the outcomes.*

11/7. If the varieties are measured logarithmically (as is almost always convenient), and if the same conditions hold, then the theorem takes a very simple form. Let V_D be the variety of *D,* V_R that of *R,* and V_O that of the outcome (all measured logarithmically).

Then the previous section has proved that V_O cannot be less, numerically, than the value of $V_D - V_R$. Thus V_O's minimum is $V_D - V_R$.

If V_D is given and fixed, $V_D - V_R$ can be lessened only by a corresponding increase in V_R. Thus *the variety in the outcomes,* if minimal, *can be decreased further only by a corresponding increase in that of R.* (A more general statement is given in S.11/9.)

This is the law of Requisite Variety. To put it more picturesquely: *only variety in R can force down the variety due to D;* **variety can destroy variety.**

This thesis is so fundamental in the general theory of regulation that I shall give some further illustrations and proofs before turning to consider its actual application.

11/8. (This section can be omitted at first reading.) The law is of very general applicability, and by no means just a trivial outcome of the tabular form. To show that this is so, what is essentially the same theorem will be proved in the case when the variety is spread out in time and the fluctuation incessant—the case specially considered by Shannon. (The notation and concepts in this section are those of Shannon's book.)

Let *D, R,* and *E* be three variables, such that each is an information source, though "source" here is not to imply that they are acting independently. Without any regard for how they are related causally, a variety of entropies can be calculated, or measured empirically. There is *H(D,R,E),* the entropy of the vector that has the three as components; there is $H_D\,(E)$, the uncertainty in *E* when *D, S* state is known; there is $H_{ED}\,(R)$, the uncertainty in *R* when both *E* and *D* are known; and so on.

The condition introduced in S.11/5 (that no element shall occur twice in a column) here corresponds to the condition that if *R* is fixed, or given, the entropy of *E* (corresponding to that of the outcome) is not to be less than that of *D,* i.e.

$$H_R\,(E) \geq H_R\,(D)$$

Now whatever the causal or other relations between *D*, *R* and *E*, algebraic necessity requires that their entropies must be related so that

$$H(D) + H_D\,(R) = H(R) + H_R\,(D)$$

for each side of the equation equals *H(R,D).* Substitute $H_R(E)$ for $H_R(D)$, and we get

$$\begin{aligned} H(D) + H_D\,(R) &\leq H(R) + H_R\,(E) \\ &\leq H(R,E). \end{aligned}$$

But always, by algebraic necessity,

$$H(R, E) \leq H(R) + H(E)$$

so $$H(D) + H_D(R) \leq H(R) + H_R(E)$$

i. e. $$H(E) \geq H(D) + H_D(E) - H(R).$$

Thus the entropy of the *E's* has a certain minimum. If this minimum is to be affected by a relation between the *D*- and *R*-sources, it can be made least when $H_D(R) = 0$, i.e. when *R is a determinate function of D.* When this is so, then *H(E)'s* minimum is *H(D) – H(R),* a deduction similar to that of the previous section. It says simply that the minimal value of *E's* entropy can be forced down below that of *D* only by an equal *increase* in that of *R.*

11/9. The theorems just established can easily be modified to give a worth-while extension.

Consider the case when, even when *R* does nothing (i.e. produces the same move whatever *D* does) the variety of outcome is *less* than that of *D.* This is the case in Table 11/4/1. Thus if *R* gives the reply α to all *D's* moves, then the outcomes are *a, b* or *d*— a variety of three, less than *D's* variety of five. To get a manageable calculation, suppose that within each column each element is now repeated k times (instead of the "once only" of S.11/5). The same argument as before, modified in that *kn* rows may provide only one outcome, leads to the theorem that

$$V_O \geq V_D - \log k - \log V_R,$$

in which the varieties are measured logarithmically.

An exactly similar modification may be made to the theorem in terms of entropies, by supposing, not as in S.11/8 that

$$H_R(E) \geq H_R(D), \text{ but that}$$
$$H_R(E) \geq H_R(D) - K.$$

H(E)'s minimum then becomes

$$H(D) - K - H(R),$$

with a similar interpretation.

11/10. The law states that certain events are impossible. It is important that we should be clear as to the origin of the impossibility. Thus, what has the statement to fear from experiment?

It has nothing to do with the properties of matter. So if the law is stated in the form "No machine can …", it is not to be over-

thrown by the invention of some new device or some new electronic circuit, or the discovery of some new element. It does not even have anything to do with the properties of the machine in the general sense of Chapter 4; for it comes from the *Table,* such as that of S.11/4; this Table says simply that certain *D*-*R* combinations lead to certain outcomes, but is quite independent of whatever it is that determines the outcome. Experiments can only *provide* such tables.

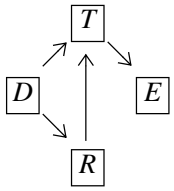
The theorem is primarily a statement about possible arrangements in a rectangular table. It says that certain types of arrangement cannot be made. It is thus no more dependent on special properties of machines than is, say, the "theorem" that four objects can be arranged to form a square while three can not. The law therefore owes nothing to experiment.

11/11. *Regulation again.* We can now take up again the subject of regulation, ignored since the beginning of this chapter, for the law of Requisite Variety enables us to apply a *measure* to regulation. I et us go back and reconsider what is meant, essentially, by "regulation".

There is first a set of disturbances *D,* that start in the world outside the organism, often far from it, and that threaten, if the regulator *R* does nothing, to drive the essential variables *E* outside their proper range of values. The values of *E* correspond to the "outcomes" of the previous sections. Of all these *E*-values only a few (η) are compatible with the organism's life, or are unobjectionable, so that the regulator *R,* to be successful, must take its value in a way so related to that of *D* that the outcome is, if possible, always within the acceptable set 17, i.e. within physiological limits. Regulation is thus related fundamentally to the game of S.11/4. Let us trace the relation in more detail.

The Table *T* is first assumed to be given. It is the hard external world, or those internal matters that the would-be regulator has to take for granted. Now starts a process. *D* takes an arbitrary value, *R* takes some value determined by *D's* value, the Table determines an outcome, and this either is or is not in η. Usually the process is repeated, as when a water-bath deals, during the day, with various disturbances. Then another value is taken by *D,* another by *R,* another outcome occurs, and this also may be either in η or not. And so on. If *R is* a well-made regulator—one that works successfully—then *R is* such a transformation of *D* that all the outcomes fall within η. *In this case R and T together are acting as the barrier F* (S.10/5.)

We can now show these relations by the diagram of immediate effects:



The arrows represent actual channels of communication. For the variety in *D* determines the variety in *R,* and that in *T* is determined by that in both *D* and *R.* If *R* and *T* are in fact actual machines, then *R* has an input from *D,* and *T* has two inputs.

(When *R* and *T* are embodied in actual machines, care must be taken that we are clear about what we are referring to. If some machine is providing the basis for *T*, it will have (by S.4/1) a set of states that occur step by step. These states, and these steps, are essentially independent of the discrete steps that we have considered to be taken by *D, R,* and *T* in this chapter. Thus, *T* gives the outcome, and any particular outcome may be compared with another, as unit with unit. Each individual outcome may, however, in another context, be analysed more finely. Thus a thirsty organism may follow trajectory 1 and get relief, or trajectory 2 and die of thirst. For some purposes the two outcomes can be treated as units, particularly if they are to be contrasted. If however we want to investigate the behaviour in more detail, we can regard trajectory 1 as composed of a sequence of states, separated by steps in time that are of quite a different order of size from those between successive regulatory acts to successive disturbances.)

We can now interpret the general phenomenon of regulation in terms of communication. If *R* does nothing, i.e. keeps to one value, then the variety in *D* threatens to go through *T* to *E,* contrary to what is wanted. It may happen that *T*, without change by *R,* will block some of the variety (S.11/9), and occasionally this blocking may give sufficient constancy at *E* for survival. More commonly, a further suppression at *E is* necessary; it can be achieved, as we saw in S.11/6, only by further variety at *R.*

We can now select a portion of the diagram, and focus attention on *R* as a transmitter:

$$D \rightarrow R \rightarrow T$$

The law of Requisite Variety says that *R's capacity as a regulator cannot exceed R's capacity as a channel of communication.*

In the form just given, the law of Requisite Variety can be shown in exact relation to Shannon's Theorem 10, which says that if noise appears in a message, the amount of noise that can be removed by a correction channel is limited to the amount of information that can be carried by that channel.

Thus, his "noise" corresponds to our "disturbance", his "correction channel" to our "regulator *R*", and his "message of entropy H" becomes, in our case, a message of entropy zero, for it is *constancy* that is to be "transmitted". Thus the use of a regulator to achieve homeostasis and the use of a correction channel to suppress noise are homologous.

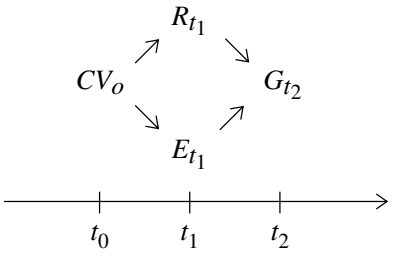
Ex. 1: A certain insect has an optic nerve of a hundred fibres, each of which can carry twenty bits per second; is this sufficient to enable it to defend itself against ten distinct dangers, each of which may, or may not, independently, be present in each second?

Ex. 2: A ship's telegraph from bridge to engine-room can determine one of nine speeds not oftener than one signal in five seconds, and the wheel can determine one of fifty rudder-positions in each second. Since experience has shown that this means of control is normally sufficient for full regulation, estimate a normal upper limit for the disturbances (gusts, traffic, shoals, etc.) that threaten the ship's safety.

Ex. 3: A general is opposed by an army of ten divisions, each of which may manoeuvre with a variety of 106 bits in each day. His intelligence comes through 10 signallers, each of whom can transmit 60 letters per minute for 8 hours in each day, in a code that transmits 2 bits per letter. Is his intelligence-channel sufficient for him to be able to achieve complete regulation?

Ex. 4: (Continued.) The general can dictate orders at 500 bits/minute for 12 hours/day. If his Intelligence were complete, would this verbal channel be sufficient for complete regulation ?

11/12. The diagram of immediate effects given in the previous section is clearly related to the formulation for "directive correlation" given by Sommerhoff, who, in his *Analytical Biology,* uses the diagram



If I am not misinterpreting him, his concepts and those used here are equivalent thus:

$$\begin{aligned} \text{Coenetic variable } (CV_o) &\leftrightarrow \text{Disturbance } (D) \\ \text{Response } (R_{t_1}) &\leftrightarrow \text{Response } (R) \\ \text{Environmental circumstances } (E_{t_1}) &\leftrightarrow \text{Table } (T) \\ \text{Subsequent occurrence } (G_{t_2}) &\leftrightarrow \text{Outcome } (E) \end{aligned}$$

A reading of his book may thus help to extend much of the theory given in this Part, for he discusses the subject extensively.

11/13. The law now enables us to see the relations existing between the various types of variety and information that affect the living organism.

A species continues to exist (S.10/14) primarily because its members can block the flow of variety (thought of as disturbance) to the gene-pattern (S.10/6), and this blockage is the species' most fundamental need. Natural selection has shown the advantage to be gained by taking a large amount of variety (as information) partly into the system (so that it does not reach the gene-pattern) and then using this information so that the flow via *R* blocks the flow through the environment *T*.

This point of view enables us to resolve what might at first seem a paradox—that the higher organisms have sensitive skins, responsive nervous systems, and often an instinct that impels them, in play or curiosity, to bring more variety to the system than is immediately necessary. Would not their chance of survival be improved by an avoidance of this variety?

The discussion in this chapter has shown that variety (whether information or disturbance) comes to the organism in two forms. There is that which threatens the survival of the gene-pattern—the direct transmission by *T* from *D* to *E*. This part must be blocked at all costs. And there is that which, while it may threaten the gene-pattern, can be transformed (or re-coded) through the regulator *R* and used to block the effect of the remainder (in *T*). This information is useful, and should (if the regulator can be provided) be made as large as possible; for, by the law of Requisite Variety, the amount of disturbance that reaches the gene-pattern can be diminished only by the amount of information so transmitted. That is the importance of the law in biology.

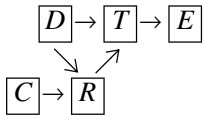
It is also of importance to us as we make our way towards the last chapter. In its elementary forms the law is intuitively obvious and hardly deserving statement. If, for instance, a press photogra-

pher would deal with twenty subjects that are (for exposure and distance) distinct, then his camera must obviously be capable of at least twenty distinct settings if all the negatives are to be brought to a uniform density and sharpness. Where the law, in its quantitative form, develops its power is when we come to consider the system in which these matters are not so obvious, and particularly when it is very large. Thus, by how much can a dictator control a country? It is commonly said that Hitler's control over Germany was total. So far as his power of regulation (in the sense of S.10/6) was concerned, the law says that his control amounted to just 1 man-power, and no more. (Whether this statement is true must be tested by the future; its chief virtue now is that it is exact and uncompromising.) Thus the law, though trite in the simple cases, can give real guidance in those cases that are much too complex to be handled by unaided intuition.

CONTROL

11/14. The formulations given in this chapter have already suggested that regulation and control are intimately related. Thus, in S.11/3, Table 11/3/1 enables *R* not only to achieve *a* as outcome in spite of all *D's* variations; but equally to achieve *b* or c at will.

We can look at the situation in another way. Suppose the decision of what outcome is to be the target is made by some controller, C, whom *R* must obey. C's decision will affect *R's* choice of α, β or γ; so the diagram of immediate effects is

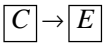


Thus the whole represents a system with two independent inputs, *C* and *D*.

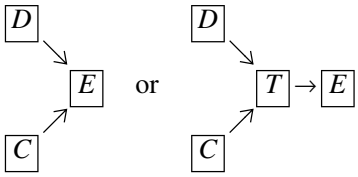
Suppose now that *R is* a perfect regulator. If C sets *a* as the target, then (through *R's* agency) *E* will take the value *a, whatever value D may take.* Similarly, if C sets *b* as target, *b* will appear as outcome whatever value *D* may take. And so on. And if C sets a particular sequence—*a, b, a. c, c, a,* say—as sequential or compound target, then that sequence will be produced, regardless of *D's* values during the sequence. (It is assumed for convenience that the components move in step.) Thus the fact that *R* is a perfect

regulator gives C complete control over the output, in spite of the entrance of disturbing effects by way of *D.* Thus, *perfect* **regulation** *of the outcome by R makes possible a complete* **control** *over the outcome by C.*

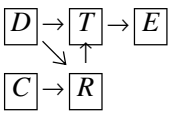
We can see the same facts from yet another point of view. If an attempt at control, by C over *E*:



is disturbed or made noisy by another, independent, input *D, so* that the connexions are



then a suitable regulator *R,* taking information from both C and *D,* and interposed between C and *T*:



may be able to form, with *T*, a compound channel to *E* that *transmits fully from C while transmitting nothing from D.*

The achievement of control may thus depend necessarily on the achievement of regulation. The two are thus intimately related.

Ex. 1: From Table 31/3/l form the set of transformations, with c as parameter, that must be used by *R* if C is to have complete control over the outcome. (Hint: What are the operands?)

Ex. 2: If, in the last diagram of this section, C wants to transmit to *E* at 20 bits/second, and a source *D* is providing noise at 5 bits/second, and *T* is such that if *R* is constant, *E* will vary at 2 bits/second, how much capacity must the channel from *D* to *R* have (at least) if C s control over *E* is to be complete?

Ex. 3: (Continued.) How much capacity (at least) is necessary along the channel from C to *R ?*

Ex. 4: (Continued.) How much along that from *R* to *T?*

11/15. In our treatment of regulation the emphasis has fallen on its property of reducing the variety in the outcome; without regulation the variety is large—with regulation it is small. The limit of this reduction is the regulation that holds the outcome rigorously constant. This point of view is undoubtedly valid, but at first it may seem to contrast sharply with the naive view that living organisms are, in general, anything but immobile. A few words, in addition to what was said in S.11/13, may be useful.

It should be appreciated that the distinction between "constant" and "varying" often depends on the exact definition of what is being referred to. Thus if a searchlight follows an aircraft accurately we may notice either that the searchlight moved through a great range of angles (angles in relation to the earth) or that the angle it made with the aircraft remained constant at zero. Obviously both points of view are valid; there is no real contradiction in this example between "great range" and "constant", for they refer to different variables.

Again, the driver who steers a car accurately from one town to another along a winding lane can be regarded either as one who has caused the steering wheel to show much activity and change or as one who, throughout the trip, has kept the distance between car and verge almost constant.

Many of the activities of living organisms permit this double aspect. On the one hand the observer can notice the great deal of actual movement and change that occurs, and on the other hand he can observe that throughout these activities, so far as they are coordinated or homeostatic, there are invariants and constancies that show the degree of regulation that is being achieved.

Many variations are possible on the same theme. Thus if variable x is always doing just the same as variable y, then the quantity $x - y$ is constant at zero. So if y's values are given by some outside factor, any regulator that acts on x so as to keep $x - y$ constant at zero is in fact forcing x to vary, copying y. Similarly, "making x do the opposite to y" corresponds to "keeping $x + y$ at some constant value". And "make the variable w change so that it is always just twice as large as v's (fluctuating) rate of change" corresponds to "keep the quantity $w - 2dv/dt$ constant".

It is a great convenience in exposition and in the processes of general theory to be able to treat all "targets" as if they were of the form "keep the outcome constant at a". The reader must, however, not be misled into thinking that the theory beat' only of immobility; he must accustom himself to interchanging the corresponding concepts freely.

SOME VARIATIONS

11/16. In S.11/4 the essential facts implied by regulation were shown as a simple rectangular table, as if it were a game between two players *D* and *R*. The reader may feel that this formulation is much too simple and that there are well known regulations that it is insufficient to represent. The formulation, however, is really much more general than it seems, and in the remaining sections of this chapter we shall examine various complications that prove, on closer examination, to be really included in the basic formulation of S.11/4.

11/17. *Compound disturbance.* The basic formulation of S.11/4 included only one source of disturbance *D,* and thus seems, at first sight, not to include all those cases, innumerable in the biological world, in which the regulation has to be conducted against several disturbances coming simultaneously by several channels. Thus, a cyclist often has to deal both with obstructions due to traffic and with disequilibrations due to gusts.

In fact, however, this case is included; for nothing in this chapter excludes the possibility that *D* may be a vector, with any number of components. A vectorial *D is* thus able to represent all such compound disturbances within the basic formulation.

11/18. *Noise.* A related case occurs when *T* is "noisy"—when *T* has an extra input that is affected by some disturbance that interferes with it. This might be the case if *T* were an electrical machine, somewhat disturbed by variations in the mains' voltage. At first sight this case seems to be not represented in the basic formulation.

It must be appreciated that *D, T*, *E*, etc. were defined in S.11/3 in purely *functional* form. Thus "*D*" is "that which disturbs". Given any real system some care may b necessary in deciding what corresponds to *D*, what to *T*, and so on. Further, a boundary drawn provisionally between *D* and *T* (and the other boundaries) may, on second thoughts, require moving. Thus one set of boundaries on the real system may give a system that purports to be of *D, T*, etc. yet does not agree with the basic formulation of S.11/4. Then it may be found that a shifting of the boundaries, to give a *new D*. *T*, etc., gives a set that *does* agree with the formulation.

If a preliminary placing of the boundaries shows that this (provisional) *T* is noisy, then the boundaries should be re-drawn so as to get *T*'s input of noise (S.9/19) included *as a component in D. D*

is now "that which disturbs", and *T* has no third input, so the formulation agrees with that of S.11/4.

There is, of course, no suggestion here that the noise, as a disturbance, can be allowed for magically by merely thinking differently about it. The suggestion is that if we start again from the beginning and re-define *D* and *T* then some *new* transformation of *D* may be able to restore regulation. The new transformation will, of course, have to be more complex than the old, for *D* will have more components.

11/19. *Initial states.* A related case occurs when *T* is some machine that shows its behaviour by a trajectory, with the outcome *E* depending on the properties of *T*'s trajectory. The outcomes will then usually be affected by which of *T*'s states is the initial one. How does *T*'s initial state come into the basic formulation of S.11/4?

If the initial state can be controlled, so that the trajectory can be started always from some standardised state, then no difficulty arises. (In this connexion the method of S.7/25 may be useful.) It may however happen, especially if the system is very large, that *T*'s initial state cannot be standardised. Does the basic formulation include this case?

It does; for *D,* as a vector, can be re-defined to include *T*'s initial state. Then the variety brought to *E* by the variety in *T*'s initial state is allotted its proper place in the formulation.

11/20. *Compound target.* It may happen that the acceptable states η at *E* may have more than one condition. Thus of a thermostat it might be demanded that

(i) it shall usually stay between 36° and 37°C;
(ii) if displaced by $\pm$ 10° it shall return to the allowed range within one minute.

This difficulty can be dealt with by the same method as in S.11/17, by recognising that *E* may be a vector, with more than one component, and that what is acceptable (η) may be given in the form of separate specifications for each component.

Thus, by allowing *E* to become a vector, the basic formulation of S. 11/4 can be made to include all cases in which the target is complex, or conditional, or qualified.

11/21. *Internal complexities.* As a last example, showing how comprehensive the basic formulation really is, consider the case in which the major problem seems to be not so much a regulation

as an interaction between several regulations. Thus a signalman may have to handle several trains coming to his section simultaneously. To handle any one by itself would be straightforward, but here the problem is the control of them as a complex whole pattern.

This case is in fact still covered by the basic formulation. For nothing in that formulation prevents the quantities or states or elements in *D, R, T,* or *E* from being made of parts, and the parts interrelated. The fact that "*D*" is a single letter in no way implies that what it represents must be internally simple or unitary.

The signalman's "disturbance" *D* is the particular set of trains arriving in some particular pattern over space and time. Other arrangements would provide other values for *D*, which must, of course, be a vector. The outcomes *E* will be various complex patterns of trains moving in relation to one another and moving away from his section. The acceptable set η will certainly include a component "no collision" and will probably include others as well. His responses *R* will include a variety of patterns of movements of signals and points. *T* is what is given—the basic matters of geography, mechanics, signalling techniques, etc., that lead determinately from the situation that has arisen and his reaction pattern to outcome.

It will be seen therefore that the basic formulation is capable, in principle, of including cases of any degree of internal complexity.